

1.1 What are the main purposes of the reviewed studies on public reporting systems?

1.1.1 System quality

System quality, in the Internet environment, measures the desired characteristics of an e-commerce system. Usability, availability, reliability, adaptability, and response time (e.g., download time) are examples of qualities that are valued by users of an e-commerce system[90].

The role of the Public Health Automated Case Event Reporting (PACER) system and FHIR interoperability standards in improving public health reporting, especially during rapid infectious disease outbreaks, has been examined [60]. Similarly, Infection Composite Scores (ICS) have been developed to help hospitals assess infection control performance over time, identify underperforming areas, and implement improvements. Furthermore, the use of data from the National Survey of Substance Abuse Treatment Services (N-SSATS) has been studied for its ability to capture additional information relevant to accessibility and service coordination [22]. Lastly, systems such as the nursing home 5-star rating system [45], quality ratings for home care institutions [48], and physician rating websites [1] have all been examined for their impact on personal healthcare decision-making.

1.1.2 Information quality

Information quality captures the e-commerce content issue. Web content should be personalized, complete, relevant, easy to understand, and secure if prospective buyers or suppliers are to initiate transactions via the Internet and return to a site on a regular basis[90].

When analyzing the rating distribution on the public reporting website Jameda, researchers examined how patient and physician characteristics influence overall scores [46]. In a related context, emerging strategies to improve vaccination rates as a quality and safety indicator for healthcare workers are also being explored [86], alongside home healthcare quality improvement efforts [89]. Moreover, studies have assessed whether performance indicators in long-term care facilities show improvement following public reporting and compared these trends with non-reported indicators [54]. Similarly, research has also examined the information valued most by referring physicians in hospital quality reports, such as hospital performance and patient experience data, to better meet physicians' needs [63]. Additionally, studies have shown that public reporting of healthcare performance data reveals disparities across insurance plans, encouraging improvement [85].

Furthermore, research on staff perceptions of publicly reported performance data highlights its role in internal decision-making and actions[72]. Additionally, studies also explore the impact of public reporting on quality improvement, particularly regarding process indicators like reperfusion rates and mortality rates for AMI patients [61], as well as the importance of quality indicators in selecting hospitals and physicians [79]. Likewise, comparative studies on evaluating provider performance [50] and the impact of public reporting and pay-for-performance incentives on quality improvement further emphasize how reputational and financial incentives drive providers to elevate care standards. The effectiveness of public reporting data in understanding and enhancing rural Australian hospitals' operational performance has also been evaluated [23]. Moreover, other studies investigate how public reporting influences physicians at various performance levels [55] and whether public reporting can improve German hospitals' quality without financial incentives [87]. Correspondingly, studies have developed systems to track critical health outcome indicators in head and neck cancer treatment to enhance clinical practice and public reporting [56].

To enhance accuracy and consistency in publicly reported infection data [57], research has explored the feasibility of using state-level electronic health record (EHR) data to measure and reduce health disparities [83]. **Studies also assess consistency in hospital recommendations in Germany's report card system and the causes of ranking inconsistencies** [64]. Evaluations of claims-based data's compliance with regulatory standards and its applications in hospital quality reporting and pay-for-performance programs have been conducted [51].

Other studies aim to understand how public reporting influences individuals' healthcare choices; for example, social media review systems [67] have significantly enhanced individuals' understanding of healthcare performance.

Additionally, studies have analyzed how different cognitive backgrounds [1] and decision-making contexts [81] affect consumer understanding and decision-making related to public reporting. Factors such as data consistency, report format (text or chart), and the use of confidence intervals have been examined for their influence on consumer comprehension and decision-making [58]. Research also examined the impact of various presentation formats, such as symbols and charts, to identify methods that improve user comprehension [65]. Studies have considered the impact of quality reports on the care of patients with advanced dementia [66], and factors affecting elderly chronic patients' perceptions of physician quality [82]. **Community studies have explored ways to use public reporting to reduce health disparities and improve awareness among minority groups [84].**

Patients often consider content such as survival and infection rates, when making choices [14]. Specifically, studies have explored how public reporting mechanisms affect general practitioners' prescribing behaviors, focusing on factors such as information availability, GP attitudes, perceived value, and perceived risk[78].

Service quality

Service quality, the overall support delivered by the service provider, applies regardless of whether the support is delivered by the IS department or a new organizational unit or is outsourced to an Internet service provider. This dimension is more important in an e-commerce environment than ever before, because the users are now customers rather than employees, and therefore, poor user support will translate into lost customers and lost sales[90].

Usage measures everything from a visit to a Web site and navigation within the site to information retrieval and execution of a transaction.

Moreover, public reporting has contributed to quality improvement in cardiovascular specialty care [62].

User satisfaction is an important means of measuring customers' opinions of an e-commerce system and should cover the entire customer experience cycle from information retrieval through purchase, payment, receipt, and service.

Additionally, some studies have focused on differences in public reporting preferences between HRC users and non-users [18].

Studies on organizational attitudes have often included physicians, as their behaviors and attitudes can represent those of the healthcare institution they work for. Studies have examined the perspectives of healthcare system stakeholders on the quality policies related to public reporting [10], as well as the attitudes of PICU directors towards benchmarking, public reporting, and pay-for-performance policies [76]. Physician attitudes toward patient experience data made public online [35], the perspectives of European infection control stakeholders on reporting HAI data [11], and the attitudes of CLC caregivers toward public reporting systems [23] have also been analyzed.

Net benefits are the most important success measures, because they capture the balance of the positive and negative impacts of e-commerce on customers, suppliers, employees, organizations, markets, industries, economies, and even society as a whole. Have Internet purchases saved individual consumers time and money? The phenomenon of "metric-driven harm" has been analyzed through case studies to examine its effects on healthcare outcomes [33], while the financial costs and administrative burdens of PCI result reporting programs on hospitals and physicians have been investigated [15].

Other studies focus on how public reporting systems have improved healthcare quality and reduced costs in Wisconsin and California [52]

Research shows that public reporting can support consumer choice and encourage providers to improve care quality [73] and that it has had positive effects on post-acute care quality in U.S. nursing homes [49], cancer surgery wait times [53], appropriate antibiotic use in URTI patients [70], and flu vaccination rates among healthcare workers [75]. Research shows that public reporting can support consumer choice and encourage

providers to improve care quality [73] and that it has had positive effects on post-acute care quality in U.S. nursing homes [49], cancer surgery wait times [53], appropriate antibiotic use in URTI patients [70], and flu vaccination rates among healthcare workers [75].

Research has also investigated the impact of public reporting on healthcare quality, particularly examining the potential positive and negative effects of report cards on the behaviors of doctors, hospitals, and patients [34].

Studies have offered practical recommendations for enhancing quality competition in Germany [17] and examined the impacts of public reporting transparency on patient choice, hospital investment, social welfare, and healthcare market structures [71] [80].

Other research analyzes transparency mechanisms, such as how the publication of prescription data affects prescribing behaviors [69]. Two public reporting programs were compared for their differing impacts on clinical and perceived quality [12].

Research also examined how public reporting and market competition influence hospitals' technical efficiency [47] and explored the relationship between patient experience and financial incentives to understand how public policy and healthcare practices can better promote patient-centered outcomes and choice [77].

Research has also investigated the impact of public reporting on healthcare quality, particularly examining the potential positive and negative effects of report cards on the behaviors of doctors, hospitals, and patients [34]. or whether mandatory public reporting assumptions effectively enhance hospital quality [59].

1.2 What are the key findings of the reviewed studies on public reporting systems?

We classified the antecedent based on the result. This led to the formation of three theme, individual related factors, organization related factors and public reporting related factors. For individual-related factors, we included evaluation tools and health literacy, while organization-related factors involved organizational attitudes, regional disparities, and adverse impacts. Public reporting-related factors focused on quality indicators and standardized frameworks. Ultimately, we categorized the studies into seven themes discussed in the following section.

System quality, in the Internet environment, measures the desired characteristics of an e-commerce system. Usability, availability, reliability, adaptability, and response time (e.g., download time) are examples of qualities that are valued by users of an e-commerce system.

Future systems should support automated, structured clinical data entry to effectively track and report treatment outcomes, thereby enhancing patient care and supporting a value-based healthcare system [56]. Significant discrepancies in monitoring practices across institutions suggest the need for standardized definitions and automated systems to improve data accuracy and consistency [57]. Standardized and validated reporting methods are also essential for improving report credibility and applicability [86]. Integration of patient experience data and real-time information is increasingly considered essential for the future of public reporting [73]. A new survey framework has been proposed to guide data collection and improve services in the future [22].

Highly rated hospitals on social media often exhibit better quality metrics, though this correlation is not absolute [67]. Low consistency in rankings among various report cards has also been observed [64].

Public reporting's influence on quality improvement has been limited, especially during the late 1990s, as third-party payers, patients, and referring physicians made little use of publicly available data for clinical decisions [74]. A unified data processing framework could address these issues by providing an effective approach to measuring and monitoring health disparities [83].

Information quality captures the e-commerce content issue. Web content should be personalized, complete, relevant, easy to understand, and secure if prospective buyers or suppliers are to initiate transactions via the Internet and return to a site on a regular basis.

The format of data presentation in public reports, such as using charts or text and including confidence intervals, does not significantly impact consumer understanding [58]. However, providing detailed explanations within public reporting data enhances individuals' ability to comprehend and make informed choices [14]. Supporting consumers in better understanding performance data is essential. Seven elements of information presentation

can improve user comprehension, such as symbol-annotated bar charts, clear labels indicating high and low values relative to hospital performance, and rankings based on performance [65].

Ratings for female doctors tend to be higher than for their male counterparts, and older patients and those with private health insurance generally provide higher ratings [46].

Future public reporting developers should focus on user experience by creating reports tailored to different user groups [81]. Specific measures can increase minority awareness of public reporting [84]. Perceptions of public reporting are influenced by factors such as educational level, health status, and trust [82]. Patients with advanced dementia tend to perform poorly on specific quality indicators, particularly those related to functional decline and weight loss [66]. MHCP patients generally score lower on quality indicators, with significant gaps persisting despite some improvements [85]. Results also indicate that users with no prior exposure to hospital report cards show a stronger intention to use them [18].

Studies suggest that hospital report cards should emphasize physician and patient experiences to better support consultations. This implies that presenting hospital quality data requires adjustments to maximize its utility in clinical decision-making [63].

However, public reporting systems may require adjustments to focus on factors patients find meaningful, such as hospital reputation and physician expertise, to increase their usage and impact [79].

Service quality, the overall support delivered by the service provider, applies regardless of whether the support is delivered by the IS department or a new organizational unit or is outsourced to an Internet service provider. This dimension is more important in an e-commerce environment than ever before, because the users are now customers rather than employees, and therefore, poor user support will translate into lost customers and lost sales.

Collaborative efforts between public health agencies and healthcare systems have enabled faster case information collection, reduced human errors, and ensured data privacy and accuracy [60].

Usage measures everything from a visit to a Web site and navigation within the site to information retrieval and execution of a transaction.

The availability of information, along with perceived value and attitudes of general practitioners, significantly impacts behavioral intentions, while perceived risk does not [78]. The availability of information and the perceived value and attitudes of general practitioners significantly influence their behavioral intentions, whereas perceived risk does not exert a significant effect [78].

User satisfaction is an important means of measuring customers' opinions of an e-commerce system and should cover the entire customer experience cycle from information retrieval through purchase, payment, receipt, and service.

Providers express concerns about how transparency may affect reputation and market share [80]. Many physicians view quality data on online resources with skepticism, doubting its accuracy. Only 2% consider Medicare's public reporting site highly accurate, and over half believe the data is not helpful for patients choosing doctors. They prefer foundational information, such as board certifications and clinical interests, over performance-based quality metrics or patient reviews [35]. While patients generally support the public reporting of performance data, they also report increased anxiety, especially when they are unable to change their healthcare provider. Many patients prefer data to be presented positively, with explanations and educational resources to help them understand and make informed decisions. Additionally, patients emphasize the importance of trusting their healthcare team and often rely more on physician advice than on statistical data alone [14].

Research suggests that physician rating websites significantly influence patients' physician selection, and this impact is likely to grow as public awareness increases [1]. The star rating system notably affects nursing homes' market shares, with consumers more likely to choose highly rated facilities [45]. Similarly, public reporting influences consumer choices, with higher-rated home care facilities being more frequently chosen after the introduction of public reporting [48].

6. Net benefits are the most important success measures, because they capture the balance of the positive and negative impacts of e-commerce on customers, suppliers, employees, organizations, markets, industries, economies, and even society as a whole. Have Internet purchases saved individual consumers time and money

policy’

Acute care hospitals in Flanders have made substantial investments in quality improvement policies, with most hospitals completing initial certifications, publicly reporting quality indicators, and participating in pay-for-performance programs. Despite this, stakeholder opinions on these policies vary [10]. Public reporting of HAI data varies significantly across European countries; experts in countries with established reporting systems generally support it, while experts in countries without such systems express reservations. Experts also suggest that reporting structure and process indicators is more beneficial for hospital quality improvement than direct infection rates, due to concerns about data misinterpretation and biases [11]. Public reporting has been shown to positively impact quality indicators and process measures, such as reperfusion rates and AMI mortality [61], and reduce antibiotic prescription rates [55], [70]. Meanwhile, CLC quality improvements [21], [88], vaccination rates [75], quality of care metrics [49], [72], infection control measures [59], [68], and cardiovascular care quality [62] have all improved following public reporting implementation.

Adverse Effects

Public reporting can also lead to unintended consequences, sometimes termed “metric-driven harm” [33]. While quality competition is credited with improving healthcare standards in Germany, quality measurement and public reporting are seen as among the most effective tools in this regard [17]. Many studies recommend that regulators prioritize incentives for hospitals to improve quality, considering interventions that encourage hospitals to enhance their services [71]. However, despite the transparency and accountability that public reporting promotes, unintended negative outcomes may arise. For instance, some physicians may avoid high-risk patients to improve their reported performance metrics. Furthermore, doctors and hospitals may focus more on meeting specific “target rates” rather than providing individualized care based on clinical judgment and patient needs. While some studies suggest that public reporting has contributed to lower cardiac surgery mortality rates, others find evidence that some physicians may refuse high-risk cases to maintain favorable evaluations. Additionally, while transparency has generally improved healthcare quality, studies reveal its impact can be limited, and in certain cases, it may even inadvertently reduce care quality [34]. Public reporting requirements for PCI have imposed a substantial financial burden on hospitals, with catheterization labs incurring annual costs between \$100,000 and \$200,000, with some spending even higher amounts [35]. Although incentive-based programs have sometimes threatened aspects of patient-centered care, combining these incentives with patient experience reports and outcomes has shown to enhance patient-valued health outcomes [77].

Evidence suggests that public reporting has had a significant impact on performance improvement [54]. Public reporting transparency policies can enhance social welfare by reducing inequalities and encouraging hospital investments in quality improvement [71]. Transparency efforts have also improved prescribing behaviors and appropriate drug use [69]. The NPC-QIC initiative has achieved significant advancements in reducing infant mortality, improving nutrition management, and optimizing care processes [80]. In Wisconsin and California, public reporting systems have significantly improved healthcare service quality while reducing costs [52]. However, while higher-performing hospitals have benefited from public reporting, lower-performing hospitals have not shown similar improvements [87]. Studies have also noted variation in improvements across institutions, with some metrics showing significant advancement while others have lagged [89], performance data from public reporting [23]. In one survey, 73% of PICU medical directors believed that benchmarking could improve patient outcomes [76]. The impact of transparency is more pronounced for younger, wealthier, or urban patients compared to rural or older patients [71]. Given the complex effects of public reporting on hospital efficiency across different market environments, policymakers are advised to consider market dynamics to achieve optimal healthcare outcomes [47]. Studies also show that while some public reporting programs have improved clinical quality in specific healthcare domains, others primarily affect public perception rather than actual quality outcomes [12]. In specific cases, such as elective surgery for lung cancer, waiting times have slightly increased [53].